

Soham KotkarSovereign Polyglot RL Evolution + KSML Alignment

Assignee: Soham Kotkar

Duration: Nov 3 – Nov 9 (4–5 focused days)

Goal (one line): Attach KSML semantic tags to every multilingual output, enable live RL policy updates from streamed MCP feedback, and publish InsightFlow telemetry — all under <1s latency on RTX 4050.

Deliverables (must-have)

1. KSML Aligner Service
 - FastAPI: POST /align.ksml → returns { text, ksml: {intent, source_lang, target_lang, karma_state, roots[]}, confidence }
 - Lightweight Sanskrit-root tagger using ksml/ksml_roots.json (lookup + heuristics).
 - Intent detector: rule+keyword + tiny classifier (optional).
2. RL Live Policy (Bandit → PPO-ready)
 - Endpoints:
 - POST /rl.feedback → {prompt, output, reward, lang, trace_id} (updates local Q-table/bandit).
 - GET /rl/policy → dump current policy snapshot.
 - Local updates (bandit/Q-table) + episode log to NAS/S3 for cloud PPO.
 - CLI: python rl/collect.py --source http://.../bridge.reason --upload s3://.../episodes/.
3. MCP Feedback Stream
 - mcp/stream_client.py pulls (prompt, target, correction) from Core/Bucket or web feeds.
 - Writes to data/feedback_stream.jsonl and feeds RL collect.
 - Supports at least 10 languages live (from your 34 set).
4. Speech-Ready Composer for Vaani

- POST /compose.speech_ready → { text, lang, tone, prosody_hint } mapped to Karthikeya's schema.
 - Confirm tone keys & test 10 samples (EN/HI + 3 Indic).
5. Multilingual Reasoning Bridge
- POST /bridge.reason → calls Bhavesh's /compose.final_text → runs KSML aligner → RL feedback → returns { final_text, ksml, prosody_hint }.
 - Logs to /logs/ksml_bridge.jsonl with trace_id, policy version, reward.
6. InsightFlow Telemetry Hooks
- Emit: latency, policy_version, reward_avg, lang_drift, cache_hit.
 - Fallback stub if InsightFlow offline.
7. Docs + Smoke
- bridge/tests/smoke_results.md across 10 languages (latency, pass/fail, ksml coverage).
 - README with curl examples, runbook, HDIG reflection.

Non-functional constraints

- Latency: end-to-end $\leq 2s$; avg per call $\leq 1s$ on RTX 4050.
- VRAM: ≤ 4 GB.
- Security: Supabase JWT check on all POSTs; no secrets in repo.
- Logs: JSONL with trace_id, user_id (if present), timestamps.
- Storage: episodes to NAS/S3 (use Yaseen/Karan credentials).
- KSML coverage: $\geq 85\%$ fields populated on smoke set.

Folder & files (exact)

sovereign_core/

```

    api.py                                # FastAPI router (all
endpoints below)
    ksm1/
        aligner.py                        # intent, karma_state, roots
        ksm1_roots.json                  # small dictionary + patterns
    rl/
        policy.py                        # bandit/Q-table (persist to
disk)
        feedback_logger.py               # writes JSONL + S3/NAS push
        collect.py                       # pulls from bridge.reason &
posts /rl.feedback
    bridge/
        bhavesh_connector.py             # calls /compose.final_text
        vaani_adapter.py                 # maps tone/prosody for TTS
    mcp/
        stream_client.py                 # streaming feedback source
        config.yml                       # endpoints (Core/Bucket/http/
hf)
    logs/
        ksm1_bridge.jsonl
requirements.txt
README.md
scripts/run_dev.sh
bridge/tests/smoke_results.md

```

Endpoints (contract)

- POST /align.ksml → { text, source_lang?, target_lang? } → { text, ksm1: {intent, karma_state, roots[]}, confidence }
- POST /compose.speech_ready → { text, lang } → { text, lang, tone, prosody_hint }
- POST /rl.feedback → { prompt, output, reward, lang, trace_id } → { ok:true, policy_version }
- GET /rl/policy → { policy_version, table_size, stats }
- POST /bridge.reason → { query, lang } → { final_text, ksm1, prosody_hint, latency_ms, trace_id }

Day-by-day plan

Day 1 — KSM1 Aligner + Bridge stub

- Implement ksml/aligner.py (rules + tiny lookup).
- Wire /align.ksml and /bridge.reason (call Bhavesh → align → log).
- Log JSONL with trace_id.
Checkpoint: 3 language smoke (EN/HI/MR).

Day 2 — RL feedback + MCP stream

- Implement rl/policy.py (bandit/Q-table), /rl.feedback, /rl/policy.
- Build mcp/stream_client.py & mcp/config.yml → stream into data/feedback_stream.jsonl.
- rl/collect.py pushes episodes to S3/NAS.
Checkpoint: 100+ episodes uploaded; policy version increments visible.

Day 3 — Vaani speech-ready + InsightFlow hooks

- Implement /compose.speech_ready and confirm tone keys with Karthikeya.
- Add telemetry emit (fallback stub) for latency, rewards, drift.
- Optimize for <1s average latency (cache, lightweight parsing).
Checkpoint: 10 voice-ready samples passed.

Day 4 — 10-language smoke + docs

- Run smoke across 10 languages; fill smoke_results.md (latency, ksml fields, reward deltas).
- README: curl, env, run_dev.sh, integration notes.
- HDIG reflection in README bottom.
Checkpoint: Open PR to Task Bank (Vinayak) with S3 paths.

Commands (copy/paste)

```
# dev
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
uvicorn sovereign_core.api:app --host 0.0.0.0 --port 9200 --
reload
```

```
# KSMML align (sample)
curl -s -X POST localhost:9200/align.ksml -H 'Content-Type:
application/json' \
  -d '{"text":"भाषा से जुड़ा प्रश्न है","source_lang":"hi"}'

# bridge → reason
curl -s -X POST localhost:9200/bridge.reason -H 'Content-
Type: application/json' \
  -d '{"query":"Explain photosynthesis briefly","lang":"en"}'

# RL feedback (reward=+1)
curl -s -X POST localhost:9200/rl.feedback -H 'Content-Type:
application/json' \
  -d '{"prompt":"...", "output":"...", "reward":1,
"lang":"en", "trace_id":"t-123"}'

# MCP stream quick test
python -c "from sovereign_core.mcp.stream_client import
stream; print(next(stream('hf_stream:hi_wiki')))"

# Episode upload
python sovereign_core/rl/collect.py --source http://
localhost:9200/bridge.reason \
  --upload s3://bhiv/rl_episodes/soham/
```

Coordination

- Bhavesh: confirm /compose.final_text contract & sample payloads.
- Karthikeya: tone/prosody field mapping for /compose.speech_ready.
- Yaseen / Karan: S3/NAS credentials for RL uploads; MCP connector endpoints.
- Vinayak (Task Bank): PR, smoke report, S3 paths validation.

Acceptance checklist

- KSMML tags present (intent, karma_state, roots) in 90%+ smoke outputs.
- RL policy updates visible and persisted; ≥ 200 episodes uploaded.
- MCP stream yields live samples for ≥ 10 languages.
- Vaani speech-ready JSON validated by Karthikeya.

- Avg latency $\leq 1s$ (4050), end-to-end $\leq 2s$.
- PR opened with README, smoke_results.md, HDIG reflection; Task Bank notified.